

Analytical Chemistry II - Titration Standards, Samples & Indicators

Compiled from the uploaded file only. Same information, larger and clearer font.

Abbreviations	St. = standard solution / titrant Sample = analyte / titrand ph.ph. = phenolphthalein M.O = methyl orange M.R = methyl red
General notes	Back titration usually needs a blank determination. If the endpoint is gradual, an indicator may not be suitable and electrochemical methods or mixed indicators are used. Weak acid/weak base titrations have no sharp endpoint.

1) Direct titrations

Experiment	Sample / analyte	Standard / titrant	Indicator	Notes
Direct titration - strong acids	HCl, HBr, HI, HClO ₄ , HNO ₃ , H ₂ SO ₄	Standard NaOH	phenolphthalein or methyl orange	Strong acids are titrated directly against standard base.
Direct titration - weak acids	Formic acid, acetic acid, chloroacetic acid, nicotinic acid	Standard NaOH	phenolphthalein (not methyl orange)	Used for weak acids; the file explicitly excludes M.O for these.
Direct titration - insoluble acid	Benzoic acid	Standard NaOH	phenolphthalein	Dissolve first in neutral ethanol, then titrate.
Direct titration - boric acid	Boric acid	Standard NaOH	phenolphthalein	Polyhydroxy compound such as glycerol is added to enhance acidity and form glyceroborate.
Direct titration - acid salts	Potassium acid phthalate, potassium acid sulphate, potassium acid tartrate	Standard NaOH	phenolphthalein	Acid salts are titrated to the ph.ph. end point.
Direct titration - strong bases	NaOH, KOH, Ba(OH) ₂	Standard acid	methyl orange or phenolphthalein	The file lists these as directly titrated bases.
Direct titration - weak bases	Weak bases; pharmaceutical bases such as aminophylline	Standard HCl	methyl orange or methyl red	phenolphthalein is not used for weak bases in this file.
Direct titration - CaO	Calcium oxide	Standard acid (used in direct titration)	phenolphthalein	CaO is made more soluble in sucrose solution and becomes alkaline to ph.ph.

2) Double indicator titrations

Experiment	Sample / analyte	Standard / titrant	Indicator	Notes
Double indicator - diprotic acids	H ₂ SO ₄ , H ₂ S, carbonic acid	Standard NaOH	M.O for 1st endpoint; ph.ph. for 2nd endpoint	Needs separate inflections; K _a ratio should be large enough for a sharp break.
Double indicator - phosphoric acid	H ₃ PO ₄	Standard NaOH	M.O for the 1st proton; ph.ph. for the 2nd proton	The 3rd proton is not determined directly because K _{a3} is very small.
Double indicator - HCl + acetic acid mixture	Mixture of HCl and CH ₃ COOH	Standard NaOH	Double indicator system	HCl is neutralized first due to common ion effect; the second endpoint is basic and lies in the ph.ph. range.
Double indicator - sodium carbonate	Na ₂ CO ₃	Standard HCl	ph.ph. for 1st endpoint; M.O for 2nd endpoint	Reaction occurs in two steps: carbonate → bicarbonate → carbonic acid.
Double indicator - carbonate / bicarbonate mixture	Na ₂ CO ₃ + NaHCO ₃ mixture	Standard HCl	ph.ph. and then M.O	Used with the same sample or a different sample of equal volume.
Double indicator - borax	Borax (Na ₂ B ₄ O ₇) and borax + boric acid mixture	Standard HCl and standard NaOH	M.O first, then ph.ph. after adding glycerol	First titration measures liberated NaOH; second titration after glycerol measures boric acid/glyceroborate.

3) Biphasic, residual/back, and indirect titrations

Experiment	Sample / analyte	Standard / titrant	Indicator	Notes
Biphasic titration - sodium salicylate	Sodium salicylate	Standard HCl	Bromophenol blue	Carry out in a separating funnel with ether. Salicylic acid must be removed from the aqueous layer immediately to avoid a false early endpoint.
Residual titration - ammonium chloride	NH ₄ Cl	Known excess standard NaOH, then back titrate residual base with standard acid	M.R. or M.O	Boil until no more NH ₃ is evolved. Blank determination is required.
Residual titration - zinc oxide	ZnO	Known excess standard HCl, then back titrate residual acid with NaOH	M.O or ph.ph.	Used because ZnO is insoluble; blank determination is required.
Residual titration - calcium carbonate	CaCO ₃	Known excess standard HCl, then back titrate residual acid with NaOH	M.O	Boil to remove CO ₂ before back titration; blank determination is required.
Residual titration - calcium oxide + calcium carbonate	Mixture of CaO and CaCO ₃	Known excess standard HCl, then back titrate residual acid with NaOH	M.O	Boil to remove CO ₂ ; blank determination is required.
Residual titration - barium chloride	BaCl ₂	Known excess standard Na ₂ CO ₃ , then back titrate residual carbonate with HCl	ph.ph.	The sample is neutral and cannot be titrated directly; blank determination is required.

Experiment	Sample / analyte	Standard / titrant	Indicator	Notes
Residual titration - aspirin	Acetylsalicylic acid (Aspirin)	Known excess standard NaOH, then back titrate residual base with HCl	phenolphthalein	Sample is refluxed with excess NaOH; blank determination is required.
Indirect titration - ammonium salts (formol method)	Ammonium chloride / ammonium salts	Formaldehyde produces HCl, then titrate with standard NaOH	phenolphthalein	This is indirect titration, not residual titration, because no known excess is added and back titrated.

Quick notes

Note	Why it matters
Blank determination	Must be done in back titration to correct titration error.
Indicator choice	Choose the indicator so its transition range overlaps the steep part of the titration curve.
Weak acid / weak base	No sharp endpoint; use mixed indicators or electrochemical methods.
Buffer zone	In weak acid-strong base or weak base-strong acid titrations, a buffer region appears before the equivalence point.